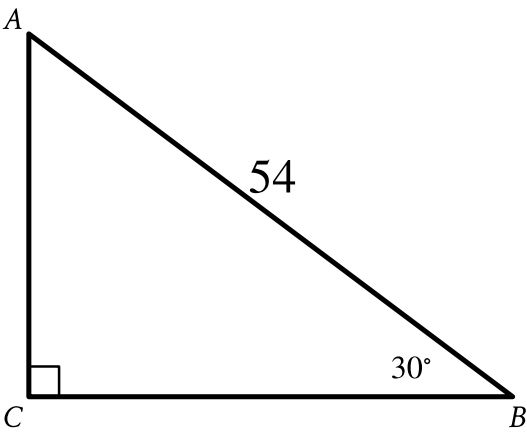


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

Right triangle ABC is shown. What is the value of $\tan A$?

Answer

- A. $\frac{\sqrt{3}}{54}$
- B. $\frac{1}{\sqrt{3}}$
- C. $\sqrt{3}$
- D. $27\sqrt{3}$

Correct Answer: C

Rationale

Choice C is correct. In the triangle shown, the measure of angle B is 30° and angle C is a right angle, which means that it has a measure of 90° . Since the sum of the angles in a triangle is equal to 180° , the measure of angle A is equal to $180^\circ - (30 + 90)^\circ$, or 60° . In a right triangle whose acute angles have measures 30° and 60° , the lengths of the legs can be represented by the expressions x , $x\sqrt{3}$, and $2x$, where x is the length of the leg opposite the angle with measure 30° , $x\sqrt{3}$ is the length of the leg opposite the angle with measure 60° , and $2x$ is the length of the hypotenuse. In the triangle shown, the hypotenuse has a length of 54 . It follows that $2x = 54$, or $x = 27$. Therefore, the length of the leg opposite angle B is 27 and the length of the leg opposite angle A is $27\sqrt{3}$. The tangent of an acute angle in a right triangle is defined as the ratio of the length of the leg opposite the angle to the length of the leg adjacent to the angle. The length of the leg opposite angle A is $27\sqrt{3}$ and the length of the leg adjacent to angle A is 27 . Therefore, the value of $\tan A$ is $\frac{27\sqrt{3}}{27}$, or $\sqrt{3}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of $\frac{1}{\tan A}$, not the value of $\tan A$.

Choice D is incorrect. This is the length of the leg opposite angle A , not the value of $\tan A$.